

The Fundamental Theorem of Calculus

A Rigorous Proof

1 Introduction

The Fundamental Theorem of Calculus (FTC) is the cornerstone of analysis: it asserts that the two principal operations of the calculus—differentiation and integration—are mutually inverse. The result is usually stated as two distinct propositions, often called Part I and Part II. Part I says that the indefinite integral of a continuous function is an antiderivative of that function; Part II says that any antiderivative may be used to evaluate a definite integral by subtraction of endpoint values.

Throughout we work with the Riemann integral on a closed bounded interval $[a, b] \subset \mathbb{R}$.

2 Preliminaries

Definition 1 (Riemann integral, briefly). A bounded function $f: [a, b] \rightarrow \mathbb{R}$ is *Riemann integrable* if its lower and upper Darboux integrals coincide; the common value is denoted $\int_a^b f(x) dx$.

We will use two standard facts without re-proof.

Lemma 2 (Boundedness on integrable intervals). *If f is Riemann integrable on $[a, b]$, then f is bounded on $[a, b]$.*

Lemma 3 (Additivity of the integral). *If f is Riemann integrable on $[a, b]$ and $a \leq c \leq b$, then f is integrable on $[a, c]$ and on $[c, b]$, and*

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx.$$

We adopt the standard convention $\int_a^a f = 0$ and $\int_b^a f = -\int_a^b f$, which makes Lemma 3 valid for any ordering of a, b, c .

3 Part I: Differentiation of the Integral

Theorem 4 (FTC, Part I). *Let $f: [a, b] \rightarrow \mathbb{R}$ be Riemann integrable, and define*

$$F(x) = \int_a^x f(t) dt, \quad x \in [a, b].$$

Then:

- (i) *F is Lipschitz continuous, hence uniformly continuous, on $[a, b]$.*

(ii) If f is continuous at a point $c \in [a, b]$, then F is differentiable at c and $F'(c) = f(c)$.

Proof. By Lemma 2, f is bounded; choose $M > 0$ with $|f(t)| \leq M$ for every $t \in [a, b]$.

Proof of (i). Let $x, y \in [a, b]$ with $x \leq y$. By Lemma 3,

$$F(y) - F(x) = \int_a^y f(t) dt - \int_a^x f(t) dt = \int_x^y f(t) dt.$$

Hence

$$|F(y) - F(x)| = \left| \int_x^y f(t) dt \right| \leq \int_x^y |f(t)| dt \leq M(y - x).$$

For arbitrary x, y we obtain $|F(y) - F(x)| \leq M|y - x|$, so F is Lipschitz with constant M , and in particular uniformly continuous.

Proof of (ii). Suppose f is continuous at $c \in [a, b]$. Fix $\varepsilon > 0$. By continuity there exists $\delta > 0$ such that

$$t \in [a, b], |t - c| < \delta \implies |f(t) - f(c)| < \varepsilon.$$

Take any h with $0 < |h| < \delta$ and $c + h \in [a, b]$. Using $\int_c^{c+h} f(c) dt = hf(c)$, we compute

$$\begin{aligned} \frac{F(c+h) - F(c)}{h} - f(c) &= \frac{1}{h} \int_c^{c+h} f(t) dt - \frac{1}{h} \int_c^{c+h} f(c) dt \\ &= \frac{1}{h} \int_c^{c+h} (f(t) - f(c)) dt. \end{aligned}$$

For every t between c and $c + h$ we have $|t - c| \leq |h| < \delta$, so $|f(t) - f(c)| < \varepsilon$. Therefore

$$\left| \frac{F(c+h) - F(c)}{h} - f(c) \right| \leq \frac{1}{|h|} \varepsilon |h| = \varepsilon.$$

The absolute-value bound holds for both signs of h because of the sign convention on the integral. Since $\varepsilon > 0$ was arbitrary,

$$\lim_{h \rightarrow 0} \frac{F(c+h) - F(c)}{h} = f(c),$$

which is precisely the assertion $F'(c) = f(c)$. □

Corollary 5. Every continuous function $f: [a, b] \rightarrow \mathbb{R}$ has an antiderivative; namely, $F(x) = \int_a^x f(t) dt$.

4 Part II: Evaluation of Definite Integrals

We first record a tool from differential calculus.

Lemma 6 (Mean Value Theorem). If $G: [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b) , then there exists $\xi \in (a, b)$ with $G(b) - G(a) = G'(\xi)(b - a)$.

Theorem 7 (FTC, Part II). Let $f: [a, b] \rightarrow \mathbb{R}$ be Riemann integrable, and suppose $G: [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$, differentiable on (a, b) , and satisfies $G'(x) = f(x)$ for every $x \in (a, b)$. Then

$$\int_a^b f(x) dx = G(b) - G(a).$$

Proof. Let $\varepsilon > 0$. Since f is integrable, there exists a partition

$$P: \quad a = x_0 < x_1 < \cdots < x_n = b$$

such that the upper and lower Darboux sums satisfy $U(f, P) - L(f, P) < \varepsilon$. On each subinterval $[x_{k-1}, x_k]$ the function G satisfies the hypotheses of the Mean Value Theorem (Lemma 6); hence there exists $\xi_k \in (x_{k-1}, x_k)$ with

$$G(x_k) - G(x_{k-1}) = G'(\xi_k)(x_k - x_{k-1}) = f(\xi_k)(x_k - x_{k-1}).$$

Telescoping the left-hand side over $k = 1, \dots, n$ gives

$$G(b) - G(a) = \sum_{k=1}^n [G(x_k) - G(x_{k-1})] = \sum_{k=1}^n f(\xi_k)(x_k - x_{k-1}).$$

The right-hand sum is a Riemann sum $S(f, P, \xi)$ for f over the partition P with tags $\xi = (\xi_1, \dots, \xi_n)$. By definition of the Darboux sums,

$$L(f, P) \leq S(f, P, \xi) \leq U(f, P).$$

The integral $\int_a^b f$ also lies between $L(f, P)$ and $U(f, P)$. Both $G(b) - G(a)$ and $\int_a^b f(x) dx$ therefore lie in the same interval of length $U(f, P) - L(f, P) < \varepsilon$, so

$$\left| (G(b) - G(a)) - \int_a^b f(x) dx \right| \leq U(f, P) - L(f, P) < \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, the difference is zero:

$$G(b) - G(a) = \int_a^b f(x) dx. \quad \square$$

Remark. The proof of Part II does *not* require f to be continuous, only Riemann integrable. The continuity of f is used in Part I to obtain $F' = f$; once an antiderivative G is given, integrability of f is enough.

5 Synthesis

Parts I and II together express the inverse relationship between differentiation and integration. Part I says: if you start with a continuous f , integrate it to get F , and then differentiate, you recover f . Part II says: if you start with a differentiable G whose derivative is integrable, then integrating that derivative recovers the net change $G(b) - G(a)$. Symbolically:

$$\frac{d}{dx} \int_a^x f(t) dt = f(x), \quad \int_a^b G'(x) dx = G(b) - G(a).$$

Corollary 8 (Integration by parts). *If $u, v: [a, b] \rightarrow \mathbb{R}$ are continuously differentiable, then*

$$\int_a^b u(x)v'(x) dx = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x) dx.$$

Proof. Apply Theorem 7 to $G(x) = u(x)v(x)$, whose derivative $G'(x) = u'(x)v(x) + u(x)v'(x)$ is continuous, hence integrable. $\square$

Corollary 9 (Substitution rule). *Let $\varphi: [a, b] \rightarrow \mathbb{R}$ be continuously differentiable with $\varphi([a, b]) \subseteq [c, d]$, and let $f: [c, d] \rightarrow \mathbb{R}$ be continuous. Then*

$$\int_a^b f(\varphi(x)) \varphi'(x) dx = \int_{\varphi(a)}^{\varphi(b)} f(u) du.$$

Proof. By Corollary 5, f has an antiderivative F on $[c, d]$. The chain rule gives $(F \circ \varphi)'(x) = f(\varphi(x)) \varphi'(x)$, which is continuous on $[a, b]$. Theorem 7 applied to $F \circ \varphi$ on $[a, b]$ and to F on $[\varphi(a), \varphi(b)]$ yields, respectively,

$$\int_a^b f(\varphi(x)) \varphi'(x) dx = F(\varphi(b)) - F(\varphi(a)) = \int_{\varphi(a)}^{\varphi(b)} f(u) du. \quad \square$$